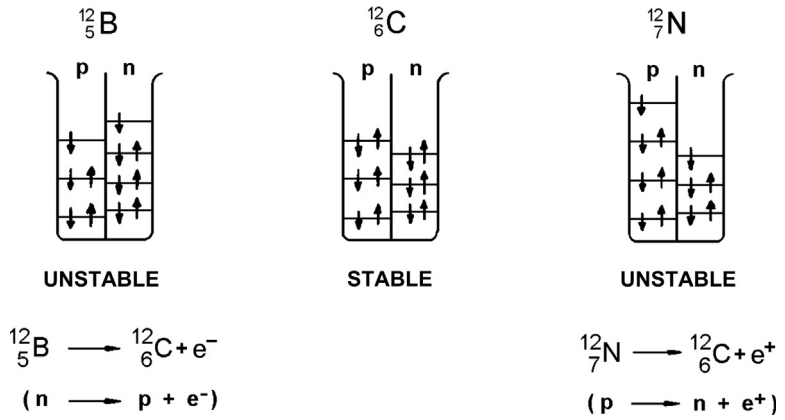


Figure 4.2 The separation and pairing of nucleons in assumed energy levels within the isobar $A = 12$. Half-life for the unstable ^{12}B is 0.02 s, and for ^{12}N 0.01 s.

develop between the last (highest energy) neutron orbital filled and the last proton orbital filled. The stability of the nucleus can be enhanced when an odd neutron in the highest neutron orbital is transformed into a proton fitting into a vacant lower energy proton orbital; see the example for $A = 12$ in Figure 4.2. In short, if the different "energy ladders" of the neutron and protons in the nucleus is investigated it becomes clear that the distance between the "steps" in the proton ladder is larger due to the repulsion force. Thus in order to achieve the lowest possible energy with a constant number of nucleons the "neuton ladder" will fill up faster thus making N/Z increase with the number of nucleons in the nucleus. These questions of nuclear forces and the energy levels of nucleons are discussed more extensively in Chapter 6.

4.3. MASS DEFECT

It was noted in Chapter 1 that the masses of nuclei (in u) are close to the mass number A. Using the mass of carbon-12 as the basis (^{1}C $\equiv$ 12 u), the hydrogen atom and the neutron do not have exactly unit masses. We would expect that the mass M_A of an atom with mass number A would be given by the number of protons (Z) times the mass of the hydrogen atom ($\mathbf{M}_\mathrm{H}$) plus the number of neutrons (N) times the mass of the neutron ($\mathbf{M}_n$), i.e.

$$M_A \approx Z\mathbf{M}_\mathrm{H} + N\mathbf{M}_\mathrm{n} \tag{4.3}$$

For deuterium with one neutron and one proton in the nucleus, we would then anticipate an atomic mass of

$$\mathbf{M}_\mathrm{H} + \mathbf{M}_\mathrm{n} = 1.007\,825 + 1.008\,665 \;=\; 2.016\,490 \text{ u} \tag{4.4}$$

When the mass of the deuterium atom is measured, it is found to be 2.014 102 u. The difference between the measured and calculated mass values, which in the case of deuterium equals −0.002 388 u, is called the *mass defect* (ΔM_A):

$$\Delta M_A = M_A - Z\mathbf{M}_{\mathrm{H}} - N\mathbf{M}_{\mathrm{n}} \tag{4.5}$$

From the Einstein equation, $E = m\mathbf{c}^2$, which is discussed further in Chapters 5 and 10, one can calculate that one atomic mass unit is equivalent to 931.5 MeV, where MeV is a million electron volts.

$$E = m\mathbf{c}^2 = 931.5\ \Delta M_A \tag{4.6}$$

where 931.5 is the mass energy content (in MeV) in 1u of mass. The relationship of energy and mass would indicate that in the formation of deuterium by the combination of a proton and neutron, the mass defect of 0.002 388 u would be observed as the liberation of an equivalent amount of energy, i.e. $931.5 \times 0.002\ 388 = 2.224$ MeV. Indeed, the emission of this amount of energy (in the form of γ-rays) is observed when a proton captures a low energy neutron to form H. As a matter of fact, in this particular case, the energy liberated in the formation of deuterium has been used in the reverse calculation to obtain the mass of the neutron since it is not possible to determine directly the mass of the free neutron. With the definition (4.5) all stable nuclei are found to have negative ΔM_A values; thus the term "defect".

In many nuclide (or isotope) tables the neutral atomic mass is not always given, but instead the *mass excess* (often, unfortunately, also called mass defect). We indicate this as δ_A and define it as the difference between the measured mass and the mass number of the particular atom:

$$\delta_A = M_A - A \tag{4.7}$$

Mass excess values are either given in u (or, more commonly, in micro mass units, μu) or in eV (usually keV). On of the advantages of using the mass excess is when calculating the energetics for nuclear reactions. In many cases the A will cancel out and thus it is possible to avoid comparing very large and very small numbers and thus increasing the accuracy of the calculations. However, with the computer power existing today this is rarely an issue. Table 4.1 contains a number of atomic masses, mass excess, and mass defect values, as well as some other information which is discussed in later sections.

When two elements form a compound in a chemical system, the amount of heat liberated is a measure of the stability of the compound. The greater this heat of formation (enthalpy, ΔH) the greater the stability of the compound. When carbon is combined with oxygen to form CO_2, it is found experimentally that 393 kJ of heat is evolved per mole of CO_2 formed. If we use the Einstein relationship, we can calculate that this would correspond to a total mass loss of 4.4×10^{-9} g for each mole of CO_2 formed (44 g).

Table 4.1 Atomic masses and binding energies

Element	Z	N	A	Atomic mass M_A (u)	Mass excess $M_A - A$ (μu)	Mass defect ΔM_A (μu)	Binding energy E_B (MeV)	E_B/A (MeV/A)
n	0	1	1	1.008665	8665	0	—	—
H	1	0	1	1.007825	7825	0	—	—
D	1	1	2	2.014102	14102	−2388	2.22	1.11
T	1	2	3	3.016049	16049	−9106	8.48	2.83
He	2	1	3	3.016029	16029	−8286	7.72	2.57
He	2	2	4	4.002603	2603	−30377	28.30	7.07
He	2	4	6	6.018886	18886	−31424	29.27	4.88
Li	3	3	6	6.015121	15121	−34348	32.00	5.33
Li	3	4	7	7.016003	16003	−42132	39.25	5.61
Be	4	3	7	7.016928	16928	−40367	37.60	5.37
Be	4	5	9	9.012182	12182	−62442	58.16	6.46
Be	4	6	10	10.013534	13534	−69755	64.98	6.50
B	5	5	10	10.012937	12937	−69513	64.75	6.48
B	5	6	11	11.009305	9305	−81809	76.20	6.93
C	6	6	12	12.000000	0	−98940	92.16	7.68
N	7	7	14	14.003074	3074	−112356	104.7	7.48
O	8	8	16	15.994915	−5085	−137005	127.6	7.98
F	9	10	19	18.998403	−1597	−158671	147.8	7.78
Ne	10	10	20	19.992436	−7564	−172464	160.6	8.03
Na	11	12	23	22.989768	−10232	−200287	186.6	8.11
Mg	12	12	24	23.985042	−14958	−212837	198.3	8.26
Al	13	14	27	26.981539	−18461	−241495	225.0	8.33
Si	14	14	28	27.976927	−23073	−253932	236.5	8.45
P	15	16	31	30.973762	−26238	−282252	262.9	8.48
K	19	20	39	38.963707	−36293	−358266	333.7	8.56
Co	27	32	59	58.933198	−66802	−555355	517.3	8.77
Zr	40	54	94	93.906315	−93685	−874591	814.7	8.67
Ce	58	82	140	139.905433	−94567	−1258941	1172.7	8.38
Ta	73	108	181	180.947993	−52007	−1559045	1452.2	8.02
Hg	80	119	199	198.968254	−31746	−1688872	1573.2	7.91
Th	90	142	232	232.038051	38051	−1896619	1766.7	7.62
U	92	143	235	235.043924	43924	−1915060	1783.9	7.59
U	92	144	236	236.045563	45563	−1922087	1790.4	7.59
U	92	146	238	238.050785	50785	−1934195	1801.7	7.57
Pu	94	146	240	240.053808	53808	−1946821	1813.5	7.56

Although chemists do not doubt that this mass loss actually occurs, at present there are no instruments of sufficient sensitivity to measure such small changes.

The energy changes in nuclear reactions are much larger. This can be seen if we use the relationship between electron volts and joules (or calories) in Appendix IV, and

observe that nuclear reaction formulas and energies refer to single atoms (or molecules), while chemical reactions and equations refer to number of moles; we have:

$$1\ \text{eV/molecule} = 1.6022 \times 10^{-19} \times 6.0221 \times 10^{23} = 96.48\ \text{kJ mole}^{-1}$$
$$= 3.8268 \times 10^{-20} \times 6.0221 \times 10^{23} = 23.045\ \text{kcal mole}^{-1} \quad (4.8)$$

Thus, the formation of deuterium from a neutron and a hydrogen atom would lead to the liberation of 214.6×10^6 kJ (51.3×10^6 kcal) for each mole of deuterium atoms formed. By comparison, then, the nuclear reaction leading to the formation of deuterium is approximately half a million times more energetic than the chemical reaction leading to formation of CO_2.

It is not common practice to use mole quantities in considering nuclear reactions as the number of individual reactions under laboratory conditions is well below 6.022×10^{23}. Therefore, in nuclear science one uses the energy and mass changes involved in the reaction of individual particles and nuclei.

4.4. BINDING ENERGY

The energy liberated in the formation of CO_2 from the elements, the heat of formation, is a measure of the stability of the CO_2 molecule. The larger the heat of formation the more stable the molecule since the more energy is required to decompose the molecule into its component atoms. Similarly, the energy liberated in the formation of a nucleus from its component nucleons is a measure of the stability of that nucleus. This energy is known as the *binding energy* (E_B) and has the same significance in nuclear science as the heat of formation has in chemical thermodynamics. We have seen that the binding energy of deuterium is 2.22 MeV. The $^4_2$He nucleus is composed of 2 neutrons and 2 protons. The measured mass of the ^{4}He atom is 4.002 603 u. The mass defect is:

$$\Delta M_{He} = M_{He} - 2\,\mathbf{M}_H - 2\,\mathbf{M}_n = 4.002603 - 2 \times 1.007825 - 2 \times 1.008665$$
$$= -0.030377\ \text{u} \quad (4.9)$$

According to eqn. (4.6) the binding energy, E_B, between the nucleons in a nucleus follows the simple relation[2]

$$E_B(\text{MeV}) = -931.5\ \Delta M_A(\text{u}) \quad (4.10)$$

Thus the binding energy for ^{4}He is 28.3 MeV. It is quite unlikely that 2 neutrons and 2 protons would ever collide simultaneously to form a ^{4}He nucleus; nevertheless, this calculation is useful because it indicates that to break ^{4}He into its basic component nucleons would require at least 28.3 MeV.

[2] The minus sign indicates that the loss of mass in formation of a nucleus releases energy.